AdifySkills

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CLASS NOTES

🏗️ Module 7 · Real-World Project Sprints

DAY 23 of 25

Project Sprint 3: Finance & Operations Autopilot

"Back-office jo khud chalta hai."

⚡ AAJ HUM YEH BANAYENGE

Finance Autopilot: Invoice Approval + Monthly P&L

Do flows ek workflow mein: (1) vendor invoice → AI extraction → ₹10,000 se kam auto-approve, upar wala manager ko email par Approve/Reject → Payables sheet. (2) Har mahine ki 1 tareekh ko sheet se P&L summary email.

Gmail | AI Extractor | Sheets | Telegram

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

Back-office (finance, HR, ops) mein sabse zyada repetitive kaam — aur sabse kam competition.

🎯 Aaj ke baad aap yeh kar paoge

01 Invoice intake & AI extraction
🌍 Daakiya bill laaya, padh ke diary mein.
⚙️ Gmail Trigger (has:attachment invoice) → AI extract → Sheet.

02 Approval chains & payment reminders
🌍 ₹10k tak bhaiya, upar papa.
⚙️ Switch amount → manager / CFO approval (Slack buttons).

03 Bank/payment reconciliation basics
🌍 Doodh wale ka hisaab milana.
⚙️ Bank statement rows vs invoices → match by amount+date → mismatch list.

04 HR onboarding automation
🌍 Naye bachche ka school mein pehla din.
⚙️ Form submit → Google account, Slack invite, welcome email, task list.

05 Management dashboards
🌍 Mahine ka report card.
⚙️ Schedule month-end → aggregate → chart → email PDF.

06 Client handover & maintenance SLA
🌍 Gaadi ki servicing contract.
⚙️ SLA: response time, monthly fee, kya cover hai, kya nahi.

⏱ Class ka plan (2 ghante)

15m	75m	20m	10m

`0:00-0:15` **Brief & Architecture**
🪄 *Mummy ki diary mein har kharcha likha hota hai aur mahine ke end mein total. Robot yahi diary apne aap bharta hai.*

`0:15-1:30` **Build Sprint**
🪄 *Marathon ek saath nahi daudte — har kilometer par paani peete hain. Har phase ek kilometer hai.*

`1:30-1:50` **Demo & Review**
🪄 *Science fair — model dikhao, batao kaise chalta hai, sawaalon ka jawab do.*

`1:50-2:00` **Capstone brief**
Capstone topics finalize
🪄 *Project ka agla page — ghar par likhna hai, kal sabko dikhana hai.*

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01
Approval chain
Amount ke hisaab se manager → finance → CEO.
🪄 **Analogy:** Leave application flow

CONCEPT 02
Reminders
Due date se pehle/baad automatic nudges.
🪄 **Analogy:** Kisi ka udhaar yaad dilana

CONCEPT 03

Reconciliation

Bank/payment entries ko invoices se match.

🪄 **Analogy:** Hisaab milana

CONCEPT 04

SLA & maintenance

Delivery ke baad support ka agreement.

🪄 **Analogy:** AMC (annual maintenance)

🔑 Aaj ka main idea

Finance/Ops autopilot = invoices, approvals, reminders, reconciliation aur reports ko jodna. Rule-based steps + AI extraction + human approval jahan paisa jaaye.

- Vendor invoice → manager approval → payment reminder
- Naye employee ka onboarding: email, Slack, docs
- Monthly P&L summary CEO ko

📖 Ek kahani se samjho: Ghar ka hisaab-kitaab robot

1 — 📨 **Bill aaye email mein** = Invoice intake

2 — 🔍 **Robot ne padha** = AI extraction

3 — 👨 **Papa ne approve kiya** = Approval chain

4 — ⏰ **Due date se pehle yaad** = Payment reminders

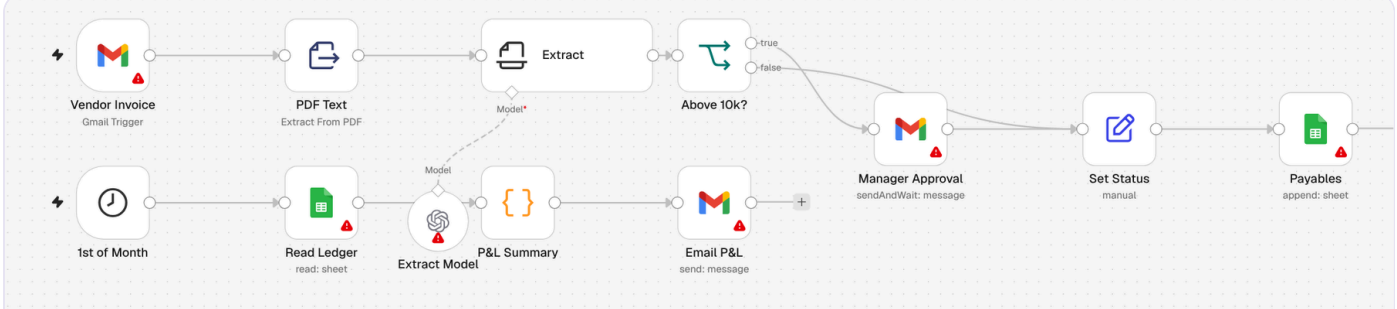
5 — 📊 **Mahine ka report card** = P&L dashboard

🪄 Ek line mein: "Mummy ki diary mein har kharcha likha hota hai aur mahine ke end mein total. Robot yahi diary apne aap bharta hai."

🗺️ Aaj ka workflow: Finance Autopilot: Invoice Approval + Monthly P&L

Data left se right chalta hai. Har dabba ek n8n node hai.





📸 Asli n8n mein import karne par yeh dikhega. Laal ⚠️ = account (credential) connect karna baaki.

💰 Isse kitna kama sakte ho?

SMEs, startups (finance autopilot) — aam taur par **₹40,000 – ₹1,00,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1. ⚡ **Gmail Trigger**
   Vendor invoice emails
2. ⚡ **Information Extractor**
   Invoice fields
3. 🔀 **Switch**
   Amount < 10k auto, 10k–1L manager, > 1L CFO
4. 💼 **Slack**
   Approval request
5. ⚡ **Schedule Trigger**
   Daily due-date reminders
6. 📊 **Google Sheets**
   Monthly P&L summary + email

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- ☐ Connect: **Gmail (Google OAuth)**
- ☐ Connect: **OpenAI API key**
- ☐ Connect: **Google Sheets (Google OAuth)**
- ☐ Sheet tabs: "Payables" (vendor, invoice_no, total, due_date, status) aur "Ledger" (date, type, category, amount)
- ☐ Manager / owner emails badlo

✅ Test kaise karein

1. ₹5,000 aur ₹25,000 ke do sample invoices email karo
2. Bade wale par approval email aana chahiye
3. P&L flow ko manually execute karke dekho

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😫 Pehle (manual)

Accountant ka roz ka data entry + follow-up — **240 min**

240 min

→

⚡ Ab (n8n)

Exceptions check karna bas — **30 min**

30 min

⏱ Accountant ka ~85% time analysis par, entry par nahi.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo 📋 Select karke copy karo

```
{{ $json.total < 10000 ? "auto" : $json.total ⩽ 100000 ? "manager" : "cfo" }}
```

📦 Sample data — input se output tak

📥 Input 📋 Select karke copy karo

```
{
  "vendor": "Cloud Hosting Co",
  "total": 45000,
  "due": "2026-12-10"
}
```

📤 Output 📋 Select karke copy karo

```
{
  "route": "manager",
  "approved": true,
  "reminder": "2026-12-08"
}
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Do flows ek workflow mein: (1) vendor invoice → AI extraction → ₹10,000 se kam auto-approve, upar wala manager ko email par Approve/Reject → Payables sheet. (2) Har mahine ki 1 tareekh ko sheet se P&L summary email.

Flow: Gmail Trigger (Vendor invoice) → PDF + AI extract (Amount, vendor) → IF (₹10k se upar?) → Gmail send & wait (Manager approval) → Sheets + Monthly P&L (Payables + report)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 Debug helper prompt 📋 Select karke copy karo

```
I'm building the n8n workflow "Finance Autopilot: Invoice Approval + Monthly P&L" (nodes: Vendor Invoice, PDF Text, Extract, Extract Model, Above 10k?, Manager Approval, Set Status, Payables, 1st of Month, Read Ledger, P&L Summary, Email P&L).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

Workflow JSON
12 nodes · ready to import
`adifyskills.com/workflows/day-23-finance-ops-autopilot.json`

Lesson page
Copy buttons, quiz aur video links
`adifyskills.com/day/23`

Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Gmail (Google OAuth)" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "OpenAI API key" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Google Sheets (Google OAuth)" connect karo.
Time par apne aap nahi chala?	"Active" toggle on hai? Settings mein timezone Asia/Kolkata check karo.
AI node par 401 / 429 error?	401 = API key galat. 429 = credit khatam ya bahut jaldi requests. OpenAI billing page check karo.
"Sheet not found" / column khaali?	Sheet ID aur tab ka naam (capital/small bhi) check karo. Pehli row mein headers hone chahiye.

⚠️ Yeh galtiyan sabse common hain

- ! Approval history save na karna (audit trail)
- ! Timezone galat — reminder raat 3 baje
- ! Client handover mein credentials ownership clear na karna

⭐ Pro tips — time bachane wali tricks

- ★ Har approval ka who/when log
- ★ Client ke apne accounts par credentials — aapke nahi
- ★ Maintenance retainer proposal ready rakho

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

Practice + Challenge

🏋️ Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Ops autopilot: invoices → extraction → approval → reminders → monthly P&L summary.

📝 Homework

Capstone project proposal submit karo (problem, solution, architecture diagram).

- [ ] Homework ho gaya

BEGINNER
Due-date reminder
- [ ]

INTERMEDIATE
Approval chain
- [ ]

ADVANCED
Payment reconciliation with fuzzy matching
- [ ]

🧩 Aaj ka challenge — pehle khud try karo

🎯 Due date se 3 din pehle vendor payment ka reminder finance team ko jaaye.

💡 Hint 1
Roz chalne wala ek alag flow chahiye jo Payables padhe.

💡 Hint 2
Schedule daily → Sheets read Payables → Filter due_date = aaj + 3 din → Gmail.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Amount ke hisaab se 3 raaste?

(A) Limit
(B) Switch
(C) Merge
(D) Sort

Q2. Daily reminders ka trigger?

(A) Form
(B) Chat
(C) Schedule Trigger
(D) Webhook

Q3. Audit ke liye kya save karein?

(A) Kuch nahi
(B) Sirf total
(C) Emoji
(D) Approval history

Q4. Delivery ke baad support agreement?

(A) SLA / maintenance
(B) SRS
(C) API
(D) SQL

📖 Aaj ke naye shabd

Trigger
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

Switch
Kai raaston mein se ek chunna — jaise traffic police kisi gaadi ko alag lane mein bhejta hai.

Schedule
Fixed time par workflow chalana — roz, har ghante, har Monday.

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✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

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⏭ Kal kya seekhenge — Day 24

Freelancing, Agency & Client Delivery

"Skill ko income mein badlo."

☐ **Day 23 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → B
Switch

Q2 → C
Schedule Trigger

Q3 → D
Approval history

Q4 → A
SLA / maintenance

✅ Challenge ka solution

Schedule (daily 10am) → Read Payables → Filter {{ $json.due_date }} equals {{ $now.plus({ days: 3 }).toFormat('yyyy-MM-dd') }} → Gmail reminder.

📚 Official docs

- Finance templates ↗
- Extract from File ↗
- Schedule Trigger ↗
- Switch node ↗
- Information Extractor ↗

▶ Videos

- n8n invoice approval workflow ↗
- n8n finance automation ↗

Doubt poochho
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wa.me/918383891716

Day 23 lesson page
Quiz, copy buttons, videos
adifyskills.com/day/23

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